



A spatio-temporal model for violence detection based on spatial and temporal attention modules and 2D CNNs

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Abstract

Violence detection is a difficult task because it involves analyzing video clips from multiple security cameras, which are located in various places and operate continuously. When violent crimes occur, a system should be able to reliably detect them in real-time and immediately alert a surveillance team. Currently, researchers employ deep learning models to detect violent behavior. Notably, a large number of deep learning approaches are based on extracting spatio-temporal information from a video by exploiting either 3D Convolutional Neural Networks (CNNs) or multi-stream networks. Despite their success, these techniques require a lot of parameters than 2D CNNs and have high computational complexity. Therefore, we present a simple spatio-temporal attention mechanism combined with a 2D CNN for an effective violence detection system. We propose a Squeeze Temporal Attention block that allows a 2D CNN to learn spatiotemporal features in videos. This effective block uses squeeze and temporal attention modules to summarize a video stream into three channels. In addition, we introduce spatial attention and feature fusion modules to improve the performance of the proposed system. The spatial attention module, Entropy Spatial Module, utilizes an entropy filter and frame differences to focus on spatial regions of the video with more movement. The fusion module parallelizes two dense layers with a 2D CNN to effectively enhance the classifier's performance. As a result, our proposed model achieves improved performance results in terms of accuracy when compared to Long Short-Term Memory, multi-stream networks, and current 3D CNNs.

Keywords Spatial attention · Temporal attention · Violence detection · Video summarization · Spatiotemporal modeling

1 Introduction

Nowadays, surveillance systems have become crucial due to frequent criminal activities at any time and place. Real-time detection of aggressive behaviors is essential for immediate police response in cases of crime. To effectively manage these events, monitoring systems must be highly efficient. The continuity of monitoring is a crucial aspect, but due to the lack of human resources, it is challenging and practically impossible for humans to continuously monitor these actions and surveillance footage without breaks. Therefore, an automatic surveillance system with violence detection capabilities can effectively perform this task.

The remarkable success of 2D Convolutional Neural Networks (2D CNNs) in the image domain has motivated researchers to explore techniques for video analysis [1, 2]. As a result, advanced deep-learning models have been proposed to mitigate the challenges associated with video analysis. Particularly, these models, such as those in research papers [3, 4], have reported superior performance compared to traditional methods [5, 6]. This progress in video analysis has opened up new research fields for applications in various domains, such as surveillance monitoring systems [7] and action recognition [8]. Detecting human actions in movies requires the consideration of both spatial information within each frame and the temporal relationships between frames [9]. Therefore, an effective deep learning architecture should be capable of encoding spatiotemporal information to achieve good performance on the task of action recognition. An extended version of 2D CNNs, 3D CNNs, is a popular method for encoding spatiotemporal information in videos [10]. These networks can extract both temporal features between consecutive frames and spatial information

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within each frame. While 3D CNNs proposed in research papers [11–14] have achieved good results in video analysis, they can be computationally expensive and require a large number of parameters. Indeed, 3D CNNs require more data to train effectively, as they have more parameters and learn spatio-temporal features from the raw video frames. Moreover, 3D CNNs are more computationally expensive than 2D CNNs, as they perform convolution operations on three dimensions instead of two. In this paper, we address these problems by proposing a new deep network architecture that combines different modules with 2D CNNs.

In addition to 3D CNNs, other methods have also been proposed for action recognition, such as those that utilize two-stream CNNs, as described in research papers [15, 16]. These methods use both RGB frames and optical flow features as input, which can provide complementary information about both the spatial and temporal characteristics of videos. However, despite their potential benefits, these methods have some limitations. One significant restriction is that two separate networks need to be trained, one for RGB frames and another for optical flow, which can be computationally expensive and time-consuming. Additionally, calculating the optical flow adds significant processing complexity, which can be challenging for real-time applications. Therefore, we present a new architecture for video analysis that can balance accuracy and efficiency while minimizing the required computational resources. Indeed, we exploit 2D CNNs that have fewer parameters than 3D CNNs in two-stream networks.

Attention mechanisms have been widely used in various machine-learning tasks, such as those reported in papers [17, 18]. These mechanisms enhance the performance of deep learning models by paying attention to informative frames and ignoring unnecessary details. In particular, attention mechanisms for video analysis combine both spatial and temporal attention modules to effectively identify crucial spatial and temporal features in video clips. Spatial attention modules are designed to highlight the most important regions within a frame, while temporal attention modules emphasize the most informative frames within video clips. By combining these two attention modules, researchers have achieved significant improvements in action recognition performance. Indeed, we propose spatial and temporal attention modules for our model. Our spatial attention module uses an entropy filter and does not increase the number of parameters. The temporal attention mechanism applies global pooling, and also has no parameters. We add two dense layers in the temporal attention module in order to make it trainable.

We present an effective method to use 2D CNNs for a violence detection system. To achieve this goal, we propose a spatial attention module that uses the entropy of two consecutive frames to detect informative regions within a video sequence. Entropy helps us find regions in

frames that are relevant to violent activities. In addition to the spatial attention module, we also exploit a channel attention module that uses a global pooling operation with dense layers to selectively emphasize the most informative channels. Furthermore, we introduce a squeeze block that compresses the input video into three channels to reduce the computational complexity of the network. The squeeze block enables 2D CNNs to analyze videos. By combining the spatial attention module with the squeeze block and the channel attention module, we provide a powerful architecture to classify violent videos based on 2D CNNs. Our proposed approach can effectively capture both spatial and temporal features of the video while significantly reducing the number of parameters. This makes the model more practical for real-time applications.

Our main contributions are summarized as follows:

- (a) Based on the more movements in violent scenes that make significant pixel differences between two consecutive frames, we propose a novel spatial attention module according to frame differences and an entropy filter. From the calculated entropy filter of frame differences, we subsequently add it to the prior frame to create spatial attention. This proposed module is named Entropy Spatial Module (ESM).
- (b) We propose a Squeeze Temporal Attention (STA) block that simultaneously squeezes a video stream into three channels and makes temporal attention. This makes it possible to use 2D CNNs to encode spatiotemporal information.
- (c) We introduce a 2D CNN architecture that includes feature fusion components to efficiently categorize input videos. The network's performance is also enhanced by these components, as demonstrated in ablation studies.

This paper is arranged as follows: Sect. 2 reviews the related works. Section 3 presents the motivation, while Sects. 4 and 5 include our proposed method and experiments, respectively. Finally, Sect. 6 gives conclusions.

2 Related works

The purpose of this section is to provide a brief overview of violence detection algorithms. These algorithms can be classified into two categories: traditional approaches and deep learning ones. While deep learning approaches offer automatic feature extraction, they may be less controllable, and they may have more difficulty producing explainable models. In contrast, the traditional methods are easier to use but require the manual extraction of complex features.

2.1 Traditional methods

Traditional methods for violence detection rely on the use of manual feature extraction techniques. In implemented systems based on the traditional method, a Support Vector Machine (SVM) is usually used as a classifier. These methods can be classified into two categories. The first category is known as local methods, and the second category is called global techniques.

Local methods are characterized by their focus on specific regions of the input video, such as individual frames or small segments of frames. They extract interest regions in a frame using techniques such as Space–Time Interest Point (STIP) [19] and Motion SIFT (MoSIFT) [20]. The primary objective of STIP and MoSIFT is to extend the concept of spatial interest points to spatiotemporal ones. As an example of local approaches for violence detection, researchers in [21] employed both MoSIFT and STIP methods to identify violent behaviors in videos. They reported that MoSIFTs outperformed STIP in terms of accuracy. Similarly, Xu et al. [22] utilized MoSIFTs, sparse coding, and Kernel Density Estimation (KDE) to identify violent behavior in a video. Specifically, the researchers used MoSIFTs and sparse coding to process selected regions in the video and extract highly discriminative features. These features were then further refined using KDE to select the most informative descriptors from MoSIFT's description.

Global techniques consider the entire video sequence as a whole. Statistical models or feature extraction techniques can be used to capture patterns of behavior and information from input videos. Moreover, these methods usually use the optical flow information of frames as an important resource to determine discriminative video features. For example, Hassner et al. proposed the Violent Flow (ViF) description based on optical flow magnitude [6]. Others had made an effort to enhance the ViF descriptor by utilizing the optical flow's magnitude and orientation. Mahmoodi et al. [23], Gao et al. [24], and Mabrouk et al. [25] proposed different descriptors that took full advantage of both the magnitude and orientation of the optical flow.

While traditional methods have obtained satisfactory results, they can be more time-consuming and labor-intensive due to the need for human expertise and creativity in designing and implementing the required features. Moreover, they can be more limited and restrictive, as the features may not capture all the information and complexity in the data, or may not be suitable for different types of problems and tasks.

2.2 Deep learning methods

Deep learning techniques have made remarkable developments and motivated researchers to exploit them for

violence detection. We categorize deep learning techniques in the violence identification field into three groups. The first category uses 3D and 2D CNNs. The second category combines different streams of deep neural networks. The third category relies on LSTM networks. We will explain these techniques in the next paragraphs.

2.2.1 Methods based on 3D CNNs and 2D CNNs

In recent years, numerous studies have focused on spatial and temporal modeling techniques to detect violence in videos. One of the significant developments in this area was the introduction of 3D CNNs. This development mitigated the limitations of 2D CNNs in encoding temporal information in videos. For example, Ding et al. [13] trained a 9-layer network on the Hockey Fight dataset, and they outperformed traditional violence detection techniques in terms of accuracy. Despite the success of 3D CNNs in tasks such as violence detection, they are computationally expensive and require many additional parameters compared to 2D CNNs. Consequently, some researchers have explored techniques to solve the inefficiencies of 3D CNNs. For instance, Song et al. [11] introduced a novel technique for interest frame extraction, whereas Mahmoodi et al. [12] suggested a block that extracted interest frames and fed them to a 3D CNN. Furthermore, other studies, including an approach proposed in [26], have focused on extracting keyframes from videos to enhance the performance of 3D CNNs. The objective of the authors in this study was to reduce the number of input frames to a convolutional network by extracting the locations of individuals and selectively attending to those frames.

Some other studies have exploited 2D CNNs instead of 3D CNNs because 2D CNNs have fewer parameters compared to 3D CNNs. For example, Keçeli et al. [27] used optical flow magnitude and orientation to create some templates and sent them to AlexNet [1] for extracting deep features. Finally, an SVM classifier was used to distinguish violent from non-violent scenes. Another method was proposed by Serrano et al. [28], who presented a technique by combining hand-crafted features with a 2D CNN. In another study, Kang et al. [29] developed a powerful approach. They introduced a new frame grouping algorithm, a spatial module, and a temporal attention algorithm to enable a 2D CNN to analyze video information. Even though these methods have obtained good performance, they usually use hand-crafted features or optical flow information. Indeed, optical flow extraction is computationally expensive and storage demanding. On the other hand, hand crafted features may not be enough or flexible, as they may miss some details and variations in data, or may not fit well with different kinds of problems and tasks.

2.2.2 Methods based on two-stream CNNs

Some violence detection systems utilize methods that involve training two or more streams simultaneously. The techniques based on two streams involve using two types of data as inputs for deep neural networks. The first input is video frames, and the second one is the optical flow of frames, which contains the pattern of motion that occurs between successive video frames. The optical flow of frames shows how objects in a video move over time. It contains the magnitude and orientation based on the pixel movements between two consecutive frames. By incorporating both video frames and optical flow information as inputs for deep networks, researchers have been able to improve the accuracy of models. Simonyan et al. [15] introduced a method in this field to recognize action in videos using a two-stream network. Xu et al. [30] used a two-stream architecture and an LSTM network to detect violent activities in videos. The authors of [31] proposed a method based on a combination of Recurrent Neural Networks (RNNs) and 2D CNNs [32] to extract both temporal and spatial features from videos. The RNNs captured the temporal features, whereas the CNNs extracted the spatial features. In addition to this, the authors also employed a novel approach by considering both video frames and their corresponding optical flow as input data for the EfficientNet [32] model. This allowed the network to learn both appearance and motion features, resulting in improved performance on their proposed model. Some other methods have employed three parallel streams for violence detection. For example, Dong et al. [33] utilized frames, optical flow, and acceleration flow as inputs for three streams. In [34], a multi-stream network was used to process handcrafted features. The authors extracted motion energy and the magnitude of the optical flow as inputs for their network. They designed an architecture that effectively detected spatial, temporal, and spatiotemporal features in videos. Another notable method presented three parallel deep networks with a soft attention mechanism [35]. In this approach, one of the branches utilized the soft attention mechanism to extract spatial features with a high probability of violent activities. Another branch took optical flow as an input, and focused on extracting temporal features, while the third branch extracted general spatial characteristics. In [36], the authors presented a novel framework for detecting anomalous events using a dual-stream convolutional neural network. Their framework contained two phases: first, they exploited a 2D CNN as an autoencoder to extract optimal spatial features from the input video frames. Second, they applied a 3D feature extraction mechanism to capture the temporal information from the video sequences.

While these methods performed well in violence detection systems, they are computationally expensive and storage-demanding, as they require pre-computing optical flow

for each video frame. Moreover, they rely on the simple linear combination of the two streams for the final fusion, which may not be optimal when one of the streams performs much better or worse than the others.

2.2.3 Methods based on 2D CNNs and LSTM

Some researchers proposed techniques based on a combination of 2D CNNs and LSTM networks. In these approaches, the 2D CNNs are used to extract spatial features from video frames, while the LSTM network is used to encode the temporal dependencies between two consecutive frames. The output of 2D CNNs is sent to the LSTM network for extracting temporal features. By combining the strengths of CNNs and LSTMs, these approaches have shown remarkable results in violence detection systems.

As an example, Abdali and Al-Tuma [37] proposed a novel method that combined the proposed network from [2] with LSTM networks. Additionally, in [38], researchers extracted spatial features from frames using 2D CNNs and a spatial transformer network, which allowed them to selectively focus on relevant regions of the video. They then fed the extracted features into an LSTM network to capture the temporal information of the video. In [39], the authors introduced a method that contained three main parts. The first part extracted features using the Xception CNN [40]. For the second and third parts, they used a Bidirectional LSTM (Bi-LSTM) along with a self-attention layer. They also introduced a dataset in their work called Surveillance Fight. Moreover, Ullah et al. [41] proposed a method that contained three steps, which were preprocessing, feature extraction, and classification. In the preprocessing step, they exploited Mask R-CNN [42] to extract interest frames. Then, they applied two parallel networks for feature extraction and an LSTM for feature learning. In [43], the authors proposed a temporal shift module and an attention block, inspired by LSTM, to detect violence in videos.

Although these methods are good at violence detection, they do not use 3D convolution on videos. This makes them less robust, and they are also slower than 3D CNNs [44].

2.2.4 Recent deep learning methods

Deep learning methods increase the power of artificial neural networks to automatically learn and extract meaningful features from input data, enabling accurate identification of violent content. In this section, we provide a detailed description of recent deep-learning methods used for violence detection. For example, researchers in [45] utilized a vision transformer. The input frames were divided into patches, which were then sent to a vision transformer unit to extract spatial features. These features were subsequently sent to an LSTM for further processing and classification.

In [46], researchers combined traditional feature extraction methods with a deep learning classifier. They first extracted features from the input videos and applied a feature selection technique to identify the most relevant ones. Subsequently, a deep neural network was employed to classify the input videos based on these selected features. In [47], researchers employed a frame differencing technique to selectively identify and extract frames with significant motion. Then, a pre-trained MobileNet model was used for classification. This technique enhanced the efficiency of computational resources. In [48], the authors proposed a semi-supervised reinforcement learning model to eliminate spatial and temporal attention modules. They exploited I3D [4] as the backbone of training procedures. In [49], a new anomaly recognition model was introduced. The authors used deep features extracted from surveillance video frames and fed them into a temporal convolution network with a multi-head attention module. In [50], the authors presented two deep models that could choose key frames from video clips. They enhanced the accuracy and stability of a violence detection system and lowered its computational complexity. Another method introduced 4D CNNs to detect violent activities in videos [51]. Residual blocks and 3D CNNs were used to learn spatiotemporal representation from videos and record inter-clip interaction. ResNet50 [52] was the backbone for the 3D CNNs, and dense optical flow was used for the region of interest.

3 Motivation

The growth of criminal events, their suddenness, and their harmful effects have forced government organizations to deploy extensive security monitoring systems to identify violent behaviors. However, automatic violence detection remains a significant challenge due to the unique characteristics and complexity of this topic. Challenges arise for automatic violence identification systems due to various factors, such as different types of violent behaviors, low-quality footage, and crowded environments. To address these challenges, we propose an effective system that incorporates different modules, such as the ESM and the STA block. In this section, we explain the motivations behind the proposed architecture.

3.1 Motivation for the ESM

We propose an ESM that focuses on spatial regions, thereby minimizing the modeling of inappropriate features in complex scenes. Specifically, the ESM is achieved by applying an entropy filter to the frame differences. After that, we use element-wise multiplication to highlight spatial regions in each frame.

3.2 Motivation for the STA block

To account for the fact that violent activities may only occur within a specific timeframe of a video clip, we employ a powerful temporal attention module that can adaptively adjust the temporal features. In contrast to using a 3D CNN, we utilize a simple 2D CNN as the backbone of our violence detection system. To enable this, we introduce a squeeze block that compresses the video streams. The block preserves the spatial dimension while reducing the temporal dimension of video clips to three, making it suitable for the input of 2D CNN. This allows the 2D CNN to model spatiotemporal information for violence detection effectively.

3.3 Motivation for the 2D CNN with feature fusion

The 3D CNNs may not be practical for some applications because of their high computational requirements and storage needs. Therefore, we utilize a 2D CNN with feature fusion for classification, which is much more efficient. To enable this, we introduce a squeeze module that is located in the STA block. The squeeze module is utilized for summarizing a video stream, which can effectively be processed by 2D CNN with feature fusion. This allows us to effectively capture spatiotemporal features in videos while minimizing computational requirements.

4 Proposed method

Our violence detection system consists of three parts: the ESM, the STA block, and a 2D CNN with feature fusion. The interaction of these modules boosts the performance of the system. Figure 1 illustrates the architecture of the proposed model. Specifically, the STA block gives the 2D CNN the ability to model spatiotemporal features. It is exploited to preserve the spatial dimension and shrink the temporal dimension of videos to three, while the ESM highlights spatial information. Additionally, the proposed 2D CNN architecture with feature fusion enhances the overall performance of the system. We go into further depth about the proposed technique in this section.

4.1 ESM

To selectively highlight spatial regions related to violent events, we propose the ESM, which focuses on moving objects. Indeed, entropy is the term used in information theory to quantify how much information is contained in data or how uncertain an event is. In image processing, entropy provides information about the structure and complexity of images. We use local entropy, which computes entropy within local regions of the image. The local entropy

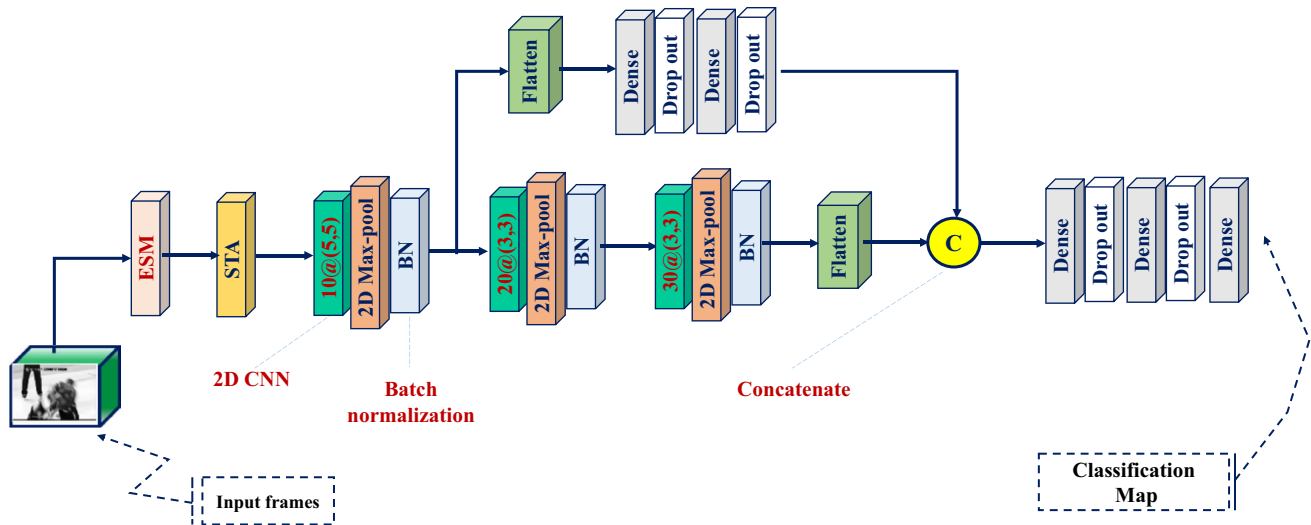


Fig. 1 The architecture of the proposed method

is calculated independently for each local region, typically using a sliding window. It provides a map of entropy values across the image and highlights regions of interest that have varying levels of complexity or texture. High local entropy in a region might indicate the presence of edges, textures, or other patterns. In our architecture, we send the frame differences to the ESM to highlight moving pixels.

The architecture of the ESM is illustrated in Fig. 2. The first step is to calculate the difference between two consecutive grayscale frames, $G_{t-1}, G_t \in R^{W \times H}$, where $1 \leq t \leq T$. Here, T is the total number of frames. Next, we apply an entropy filter to the frame differences, as shown below:

$$a_t = \text{entropyfilter}(|G_t - G_{t-1}|) \quad (1)$$

Local entropy in an image measures the complexity of the texture and patterns in a specific neighborhood of the image. The choice of a structuring element is crucial in computing local entropy. In the ESM, we use a disk-shaped structuring element with a radius of 9 pixels.

The next step is an element-wise multiplication of a_t and $|G_t - G_{t-1}|$ to highlight the moving objects. This is expressed as:

$$b_t = a_t \odot |G_t - G_{t-1}| \quad (2)$$

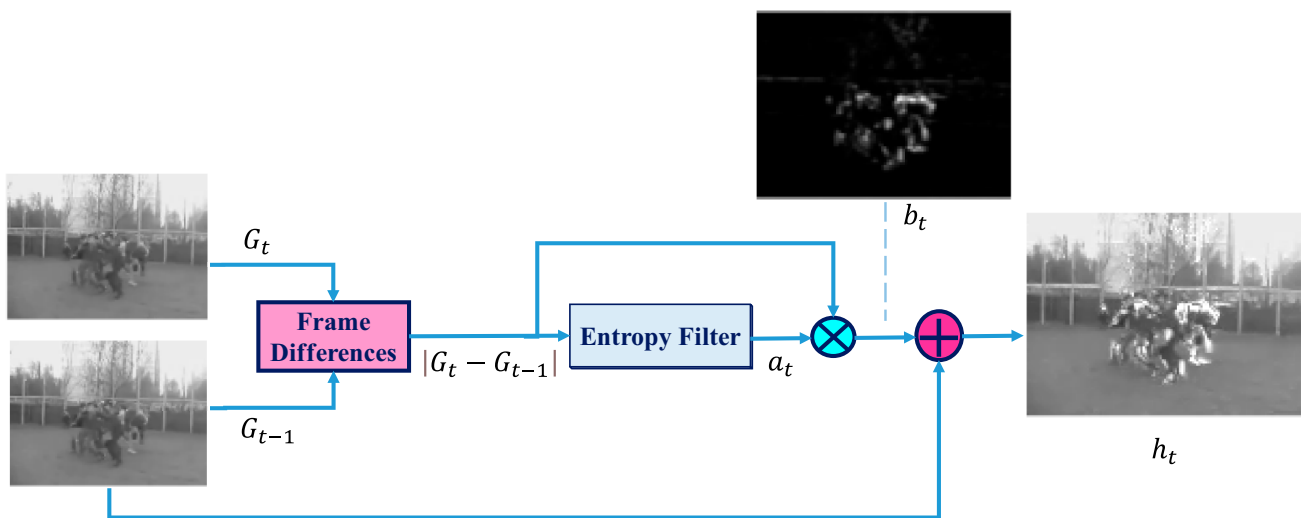


Fig. 2 Architecture of the ESM

where $\odot$ shows the element-wise multiplication. The last step of the ESM is the summation of b_t and G_{t-1} that can be shown below:

$$h_t = b_t + G_{t-1} \quad (3)$$

As shown in Fig. 2, the result images, $h_t \in R^{W \times H}$ for $1 \leq t \leq T - 1$, are generated by the ESM, while the moving objects in the video frames are highlighted. h_t are then sent to the STA block for further processing. The ESM effectively creates spatial attention for moving objects by focusing on the most informative pixels between two consecutive frames.

Figure 3 shows the visualization results for different datasets to illustrate the effect of the ESM. These results include the RGB and grayscale formats of two consecutive frames, b_t , and h_t as well as the resulting visualization of the ESM. These visualizations indicate that the ESM can effectively highlight moving objects in a scene, thus improving the accuracy of violence detection.

4.2 STA block

In our proposed system, we use a 2D CNN and introduce the STA block, which allows us to capture spatiotemporal features in videos effectively. The STA block consists of two parts: a temporal attention part and a squeeze part. The temporal attention part is designed to selectively focus on informative frames in the video sequence. In contrast, the squeeze part shrinks the temporal dimension of the video into three channels. This approach allows us to model spatio-temporal features in videos using a 2D CNN architecture, which is more efficient and practical than a 3D CNN. Figure 4 illustrates the architecture of the STA block and its two components.

We introduce the temporal attention mechanism. The input in this section is $h_t \in R^{W \times H}$ for $1 \leq t \leq T - 1$ and T is the total number of frames. Here, we randomly choose sixteen frames of h_t to improve the compatibility with [11, 12]. We denote it as $V \in R^{W \times H \times 16}$, where $W \times H$ shows spatial information. For aggregating spatial information, global average pooling has been chosen. The first dense layer consists of two neurons with ReLU activation functions, and the second one has sixteen neurons with sigmoid activation functions. Both dense layers also include dropouts. Their dropout rate is 30%. In short, the temporal attention module can be formulated as:

$$TA = \sigma(W_1 \max(0, \text{averagepool}(V)W_0)) \quad (4)$$

where σ denotes the sigmoid activation function, $W_0 \in R^{2 \times 1}$, $W_1 \in R^{16 \times 1}$. ReLU activation function is also defined as $\max(0, f(x))$, in which $f(x) = \text{averagepool}(V)W_0$. After computing the temporal attention weights (TA), we multiply them by the selected frames (V) to focus on the most informative channels. This gives us the output of the temporal attention part of the STA block, which is a tensor

of shape $W \times H \times 16$. We denote this tensor by $M \in R^{W \times H \times 16}$.

Another part of the STA block is the squeeze part, which is implemented by a simple 2D convolutional block with (1, 1) kernel size. The number of output filters in this convolutional layer is three, and its activation function is ELU. This part aims to squeeze $M \in R^{W \times H \times 16}$ into $I \in R^{W \times H \times 3}$ which is considered as the input for the next section.

For better understanding the implementation of the STA block, we choose a scenario. Consider a 1-s video clip with 24 frames per second. For every two consecutive frames, we get h_t . So, we have $h_t \in R^{W \times H}$ for $1 \leq t \leq 23$. These go to the STA block, but we first randomly pick sixteen frames of h_t to lower the processing time and deal with the hardware limitations. We named the randomly selected frames as $V \in R^{W \times H \times 16}$. Finally, we calculate the temporal attention weights from V and call the output of the temporal attention part as $M \in R^{W \times H \times 16}$. These are then squeezed to $I \in R^{W \times H \times 3}$ by the squeeze part of the STA.

4.3 2D CNN with feature fusion

In this section, we will describe the structure of a 2D CNN as depicted in Fig. 5. Our architecture consists of two branches. The first branch consists of three convolutional layers, followed by max-pooling and batch normalization layers. The first convolutional layer in our architecture utilizes a kernel size of (5 × 5), while the second and third convolutional layers employ a kernel size of (3 × 3). For all the max-pooling layers in our architecture, a stride of (2 × 2) is utilized. Additionally, all the activation functions applied to the convolutional and dense layers are ELU. The architecture incorporates a dropout rate of 20%, which helps mitigate overfitting by randomly disabling 20% of the neurons during training. In the second branch, we have a flatten layer and two dense layers. The first dense layer comprises 300 neurons, while the second layer comprises 100 neurons. Furthermore, the architecture employs a concatenate layer to merge the output of the two streams. The concatenate layer combines information from different pathways or branches in the network, enabling the model to capture and utilize diverse features or representations from multiple sources. After the concatenate layer, we have three dense layers with 300, 100, and 2 neurons, respectively. Finally, the soft-max operates as the activation function of the last dense layer.

5 Experiments

In this section, we will compare our approach to the other methods that use either handcrafted features or deep learning architectures. To begin with, we will describe four benchmark datasets that we used to evaluate our proposed method.

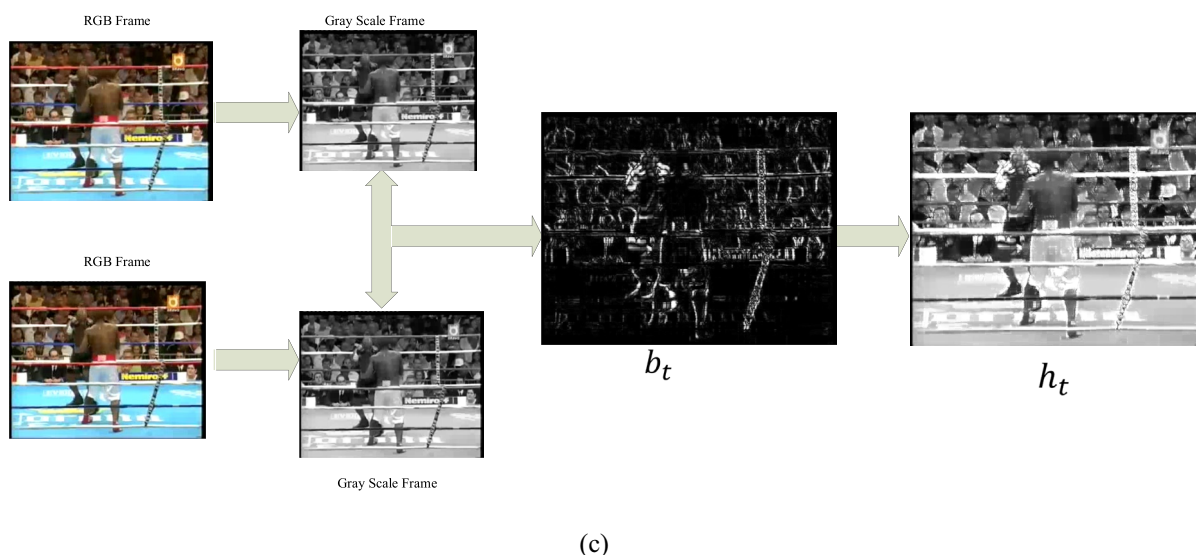
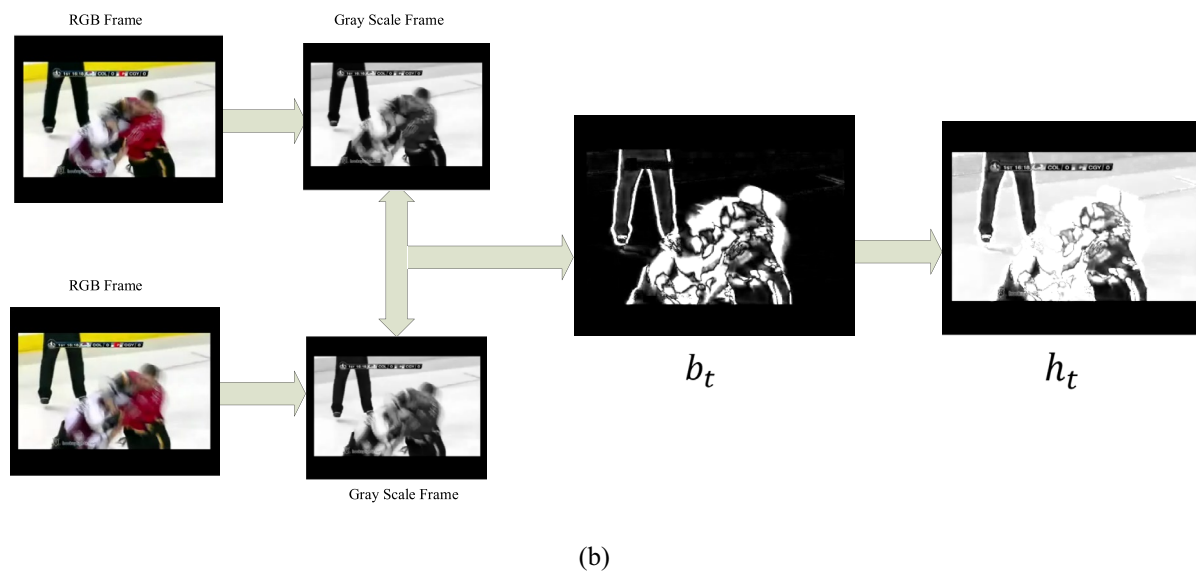
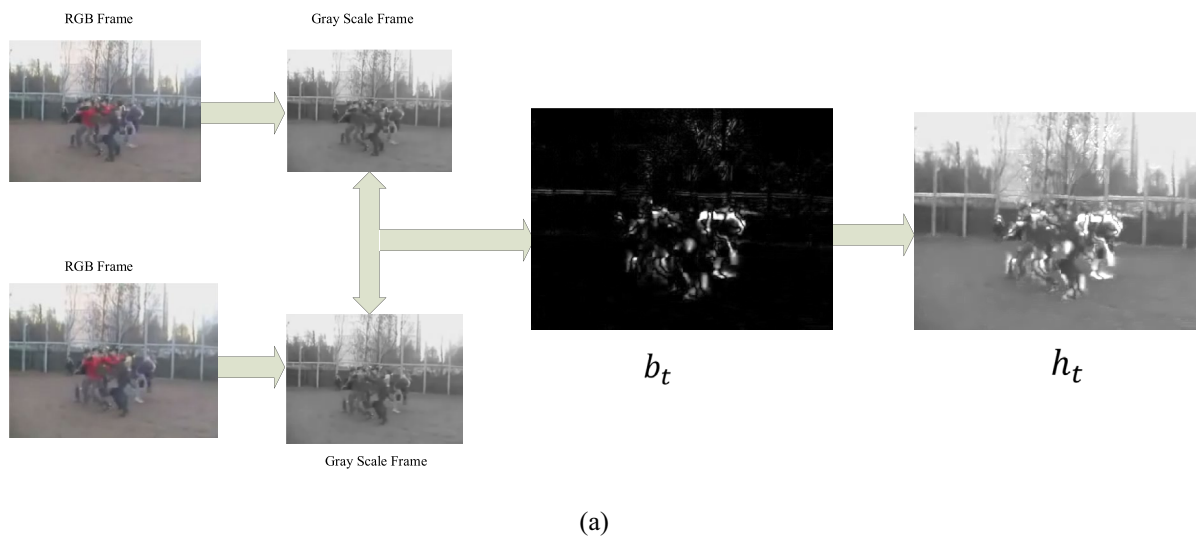


Fig. 3 Visualization results of the ESM **a** Violent Flows, **b** Hockey Fight, **c** Action Movies and **d** Surveillance Fight

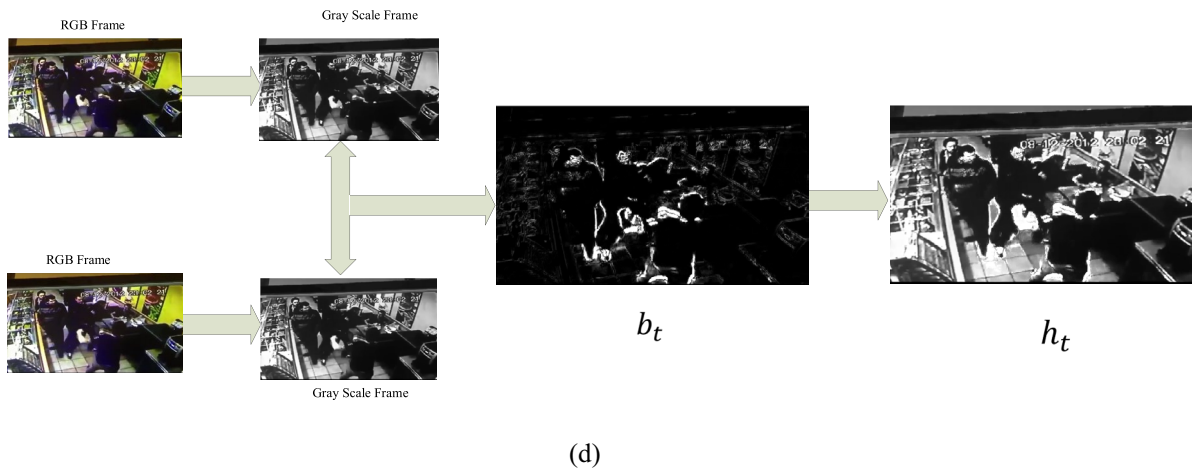


Fig. 3 (continued)

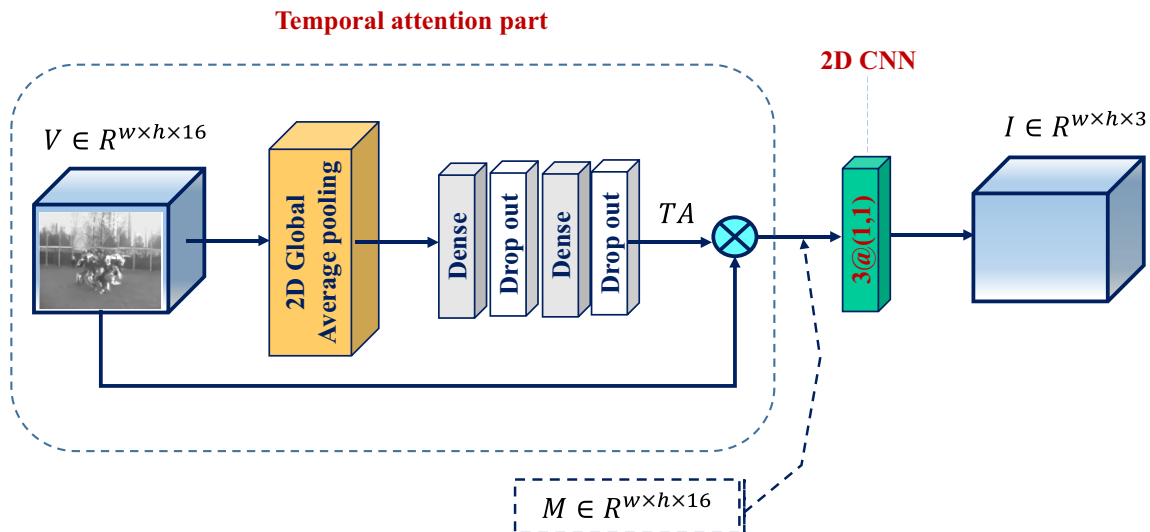


Fig. 4 The architecture of the STA block

After that, we will explain how we implemented our solution and provide a detailed comparison of the results. Additionally, we will present computational complexity and ablation studies to analyze the impact of specific components or changes in our approach. This will help us gain a better understanding of our method and how it performs compared to other approaches.

5.1 Datasets

The proposed method has been assessed and evaluated on four benchmark datasets, namely Hockey Fight [21], Action Movies [21], Surveillance Fight [39], and Violent Flows [6]. These datasets were selected to provide a diverse range of

scenarios and contexts for testing and validating the effectiveness and performance of our approach.

Figure 6 presents a collection of non-violent frames from the datasets, while Fig. 7 showcases scenes depicting violence. The frames displayed in both figures are organized into rows, each representing a specific dataset. In both figures, the first row showcases frames from the Hockey Fight dataset. The second row contains frames from the Violent Flows dataset. The third and fourth rows display frames from the Action Movies and Surveillance Fight datasets, respectively.

The Hockey Fight dataset consists of 1000 video clips of hockey game, equally split between violent and non-violent footage. There are 41 frames on average with a minimum of 40 and a maximum of 49 frames, respectively.

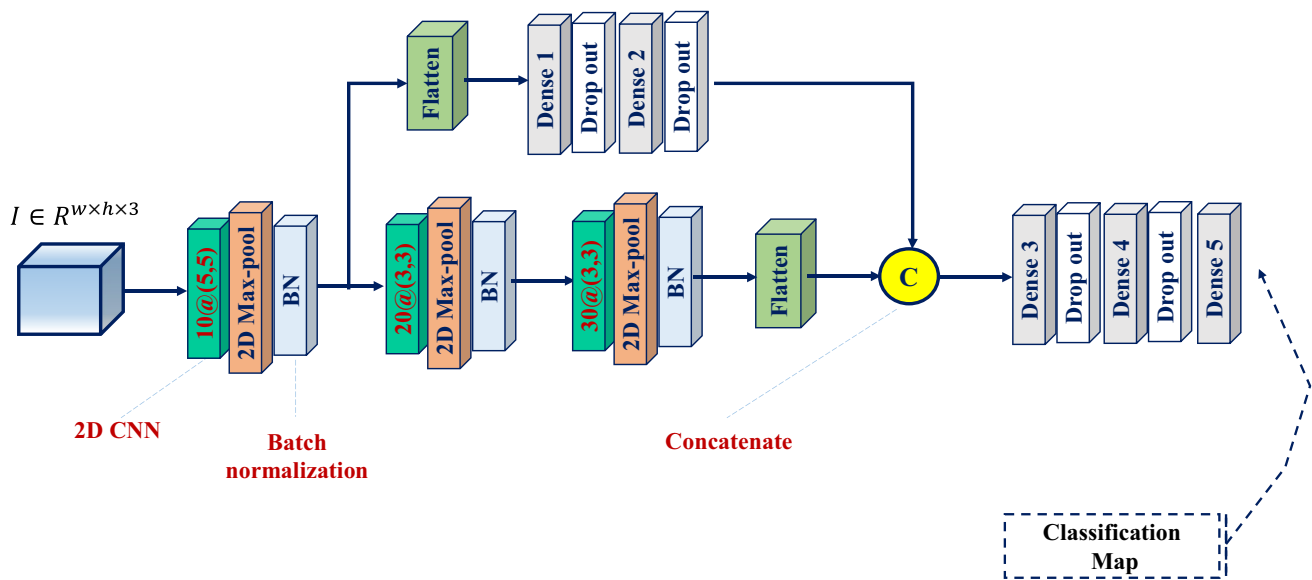


Fig. 5 The 2D CNN architecture

On the other hand, the 246 video clips from crowded environments that make up the Violent Flows dataset are equally split into violent and non-violent clips. The clips have an average of 89 frames, with a minimum of 26 and a maximum of 169 frames. The Action Movies dataset consists of 200 video clips sourced from action movies. These clips are evenly distributed between violent and non-violent situations. On average, each video clip in this dataset contains 42 frames, with a maximum of 60 frames and a minimum of 42 frames. In the Surveillance Fight dataset, 300 video clips are equally divided into fight and non-fight videos with different sizes and different numbers of frames. There are a variety of fight clips, including kicking, striking with items, and grappling. Additionally, videos are collected from different places to increase the diversity of the dataset. The dataset includes a minimum of 20 frames, a maximum of 142 frames, and an average of 57 frames. We observe that for all datasets, the video clips with violence labels contain violent activities throughout the whole video length.

The datasets used in this study are briefly described in Table 1. The table indicates that the Hockey Fight dataset has the largest number of clips (1000), while the Action Movies dataset has the smallest (200). The Violent Flows dataset has the greatest average number of frames per clip (89), while the Hockey Fight dataset has the least (41). Furthermore, the Violent Flows dataset has the widest span of frames per clip, with a low of 26 and a high of 169 frames per clip. Moreover, the frame size varies across the datasets, ranging from 240×320 to 576×720 . The Surveillance Fight dataset has different frame sizes for different videos.

5.2 Implementation details

We have developed our method using Python-Keras. The optimization algorithm is Root Mean Squared Propagation (RMSProp), with a learning rate of 0.0001. The batch size of the Violent Flows, Surveillance Fight, and Action Movies datasets is set to 48. On the other hand, the batch size of the Hockey Fight dataset is specified as 64. Google Colab is used to compile the results. To improve comparability with [11–13] and reduce processing time, the input frames are resized to (50×80) , (100×180) , (100×198) , and (60×90) for Violent Flows, Action Movies, Surveillance Fight, and Hockey Fight datasets, respectively. Table 2 provides a summary of the implementation details across four datasets. We performed the five-fold cross-validation six times and got thirty runs. We then reported the number of test and train samples in Table 2.

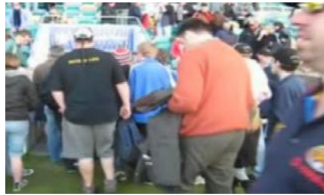
5.3 Comparison with other methods

We compare our algorithm to deep learning and traditional methods in this section. We use accuracy and standard deviation as the main metrics to measure our model's performance. We apply the five-fold cross-validation six times to verify the results' reliability and robustness. Therefore, we obtain the accuracy and standard deviation values from thirty runs of our algorithm. This method enables us to conduct a thorough evaluation and consider the possible changes in the model's performance.

We further assess our model's performance using precision, recall, F-score, and the Area Under the Curve (AUC). The corresponding results on four benchmark datasets are



a) Non-violent scenes from the Hockey Fight dataset



b) Non-violent scenes from the Violent Flows dataset



c) Non-violent scenes from the Action Movies



d) Non-violent scenes from the Surveillance Fight dataset

Fig. 6 Non-violent scenes from benchmark datasets

presented in Table 3. Additionally, the summed confusion matrix is acquired and depicted in Fig. 8. Indeed, the confusion matrix is derived from thirty independent runs across four datasets. After conducting six iterations using five-fold cross-validation, we obtained thirty confusion matrices. The reported values represent the cumulative sums within each cell of the confusion matrix.

The comparison results are presented in Table 4, demonstrating the superior performance of our suggested method in terms of accuracy. Specifically, on the Hockey Fight dataset, our approach achieved an accuracy of 99.7% with a Standard Deviation (SD) of 1.44. Similarly, on the Violent Flows dataset, we achieved an accuracy of 98.53% with a standard deviation of 6.83. Notably, the Action Movies dataset exhibited the highest accuracy of 100%. Finally, on the Surveillance Fight dataset, we obtained an accuracy of 98.44%

with a standard deviation of 6.9. These results highlight the effectiveness and reliability of our proposed method across different datasets.

Our work is compared to state-of-the-art techniques by considering both deep learning and traditional methods. Our method surpasses others in terms of accuracy, owing to three factors. Our method uses the ESM to identify spatial regions in the frames by emphasizing their spatial features. Within the ESM, the entropy filter highlights moving objects in a scene by focusing on areas with high entropy. The entropy filter distinguishes the areas with high entropy, which have more detail or texture, from the areas with low entropy, which are smoother or more uniform. By adding the entropy of frame differences to the previous frame, we can accentuate moving objects in the spatial domain. Additionally, the temporal attention module in our approach is designed to



a) Violent scenes from the Hockey Fight dataset



b) Violent scenes from the Violent Flows dataset



c) Violent scenes from the Action Movies



d) Violent scenes from the Surveillance Fight dataset

Fig. 7 Violent scenes from benchmark datasets**Table 1** A brief description of datasets

Dataset	Number of clips	Number of frames			Frame size
		Minimum	Average	Maximum	
Hockey Fight [21]	1000	40	41	49	288×360
Violent Flows [6]	246	26	89	169	240×320
Action Movies [21]	200	42	49	60	576×720
Surveillance Fight [39]	300	20	57	142	Different sizes

Table 2 Implementation details of the proposed method

Dataset	Frame size	# of test samples	# of train samples	Optimizer	Batch size
Hockey Fight	60×90	6000	24,000	RMSProp	64
Violent Flows	50×80	1476	5904	RMSProp	48
Action Movies	100×180	1200	4800	RMSProp	48
Surveillance Fight	100×198	1800	7200	RMSProp	48

Table 3 Precision, recall, F-score, and AUC results on four benchmark datasets

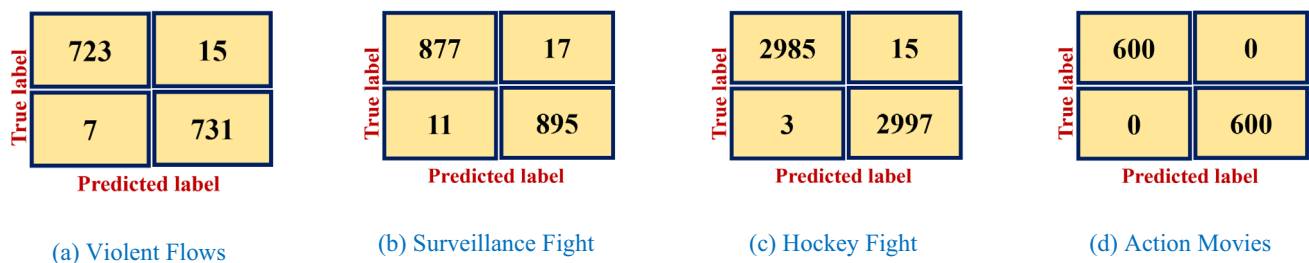
Dataset	Precision	Recall	F-score	AUC
Hockey Fight	99.5	99.9	99.7	99.95
Violent Flows	97.97	99.04	98.5	98.91
Action Movies	100	100	100	100
Surveillance Fight	98.1	98.76	98.43	98.92

capture the temporal changes and motion patterns across frames in a video. It helps to emphasize the frames that contain significant changes and reduce the influence of the frames with little or no changes. By combining both spatial and temporal attention, our approach can effectively capture both spatial and temporal information in videos, leading to superior performance compared to methods that only use one type of attention. The final component of our approach involves replacing the 3D CNNs with 2D CNNs. To accomplish this, we summarize the temporal dimension of videos

to three channels, allowing us to use the capabilities of a 2D CNN to detect violence in videos. This approach can be beneficial in scenarios where 3D CNNs are computationally expensive or where the dataset size is limited, as 2D CNNs are often more computationally efficient and require fewer parameters to train.

We compared the performance of our proposed method with traditional methods on the four benchmark datasets. Our proposed method outperformed the ViF [6] method, which only considered changes in optical flow magnitude and disregarded changes in optical flow orientation. Although other methods, such as OViF [24] and HOMO [23], considered variations in both orientation and magnitude, they were still unable to outperform our suggested strategies. These results suggest that our proposed architecture, which includes a 2D CNN, the ESM, and the STA block, can significantly improve the classifier's performance.

In addition, previous approaches such as [26, 41] utilized object detection algorithms to identify regions of interest where violent behaviors may occur. However, the

**Fig. 8** Summed confusion matrix of four datasets**Table 4** Comparison results with the state-of-the-art methods

Method	Hockey Fight		Violent Flows		Action Movies		Surveillance Fight	
	Accuracy	SD	Accuracy	SD	Accuracy	SD	Accuracy	SD
ViF [6]	81.6	0.22	81.2	1.79	96.7	0.3	–	–
OViF [24]	84.2	3.33	76.8	3.9	–	–	–	–
HOMO [23]	89.3	0.91	76.83	1.76	–	–	–	–
3D CNN + Interest frames [12]	99.4	0.73	97.49	3.32	100	–	–	–
Hough forest + 2D CNN [28]	94.6	0.6	–	–	99	0.5	–	–
Modified 3D CNN [11]	98.96	–	–	–	99.97	–	–	–
Object detection + LSTM [41]	98	–	98.21	–	–	–	74	–
Object detection + 3D CNN [26]	96	–	98	–	99.9	–	–	–
Two-cascade TSM [43]	98.995	–	97.959	–	–	–	–	–
Dual stream CNN + echo state network [36]	99	–	99.01	–	–	–	93.1	–
Vision-based fight detection [39]	98	–	–	–	100	–	72	–
Edge Vision [53]	98.5	–	–	–	–	–	75.9	–
EVOKEYNET + DEEPKEYFRM [50]	98.98	–	99.29	–	–	–	–	–
Violence 4D [51]	100	–	97.29	–	100	–	–	–
Proposed method	99.7	1.44	98.53	6.83	100	–	98.44	6.9

object detection networks had an excessive number of parameters. Despite this, our proposed solution outperformed these methods, even though they utilized architectures with many parameters to focus on regions of interest.

Furthermore, previous methods, such as a 3D CNN with interest frames [12], a modified 3D CNN [11], and DEEPKEYFRM [50] utilized techniques to focus on frames with high movement. However, despite their focus on identifying frames with potential violent behaviors, our proposed method achieved superior results. This is because our approach, which incorporates the ESM and the STA block, can more effectively encode spatiotemporal features. Our method outperformed the method proposed in [50] on the Hockey Fight dataset but underperformed on the Violent Flows dataset. We achieved an accuracy rate of 99.7% on the Hockey Fight dataset, while they obtained an accuracy rate of 98.98%. However, on the Violent Flows dataset, they had an accuracy rate of 99.29%, while we only reached an accuracy of 98.53%. In [50], the authors designed two networks for the classification task. The first one, EvoKeyNet, used an evolutionary algorithm to select features. The second one, KFCRNet, combined LSTM, Bi-LSTM, and GRU models with a voting scheme to classify input videos.

Another method [28] proposed a strategy that enables a 2D CNN to encode spatiotemporal features. However, their accuracy rate of 94.6 on the Hockey Fight dataset was low for a deep learning approach. They voted on the frames based on the probability of violent behaviors and employed the BRISK descriptor [54], the FAST detector [55], and Hough Forests. Our results demonstrate that the ESM has a better estimation of regions of interest in a frame compared to the BRISK and FAST detectors.

The authors of [43] proposed a violence detection system using a Temporal Shift Module (TSM) and a channel attention algorithm. However, our method outperformed theirs, demonstrating the robustness of our approach. Additionally, our method utilizes 2D CNNs, which reduces the number of parameters compared to the 3D CNNs used in their method. The authors of [36] utilized a dual-stream CNN framework for anomaly detection in videos. They used a 2D CNN as an autoencoder to extract spatial characteristics, which were then passed to the echo state network. While they performed well on the Surveillance Fight dataset in comparison to other researchers, they were still unable to surpass the performance of our approach. Furthermore, their accuracy rate of 99.01 on the Violent Flows dataset outperformed our model.

In [51], the authors used ResNet50 [52] and dense optical flow. They achieved an accuracy rate of 100% on the Hockey Fight dataset, which was higher than our method. However, their method, which had too many parameters because of ResNet50, was less effective than ours on the Violent Flows dataset.

According to Table 4, our proposed method was not as effective as the method proposed in [36] on the Violent Flows dataset. This dataset is used to analyze violence detection in crowded scenes and includes real-world clips of crowd violence, such as fights, riots, and sports clashes. It is a challenging dataset due to the presence of various types of violence, occlusion, noise, illumination changes, and complex scenes. Therefore, methods that use multi-stream architectures are more likely to outperform our method in terms of accuracy on this dataset.

5.4 Computational complexity

This section presents the computational complexity of our method, which uses 2D CNNs. We compare our method's complexity with the methods that rely on 3D CNNs as the main module.

The output of a 3D CNN can be written as below:

$$O_{x,y,z}^{l,n} = f \left(\sum_d \sum_{a=0}^{a_l-1} \sum_{c=0}^{c_l-1} \sum_{e=0}^{e_l-1} O_{x+a,y+b,z+e}^{l-1,d} * w_{a,c,e}^{l,n,d} + b^{l,n} \right) \quad (5)$$

where n represents the number of feature maps in layer l . $O_{x,y,z}^{l,n}$ is the output of 3D convolution at position x, y, z , and $b^{l,n}$ is the network bias. The activation function is shown as f . The set of output feature maps of layer $l-1$ is considered as d which is the input of layer l . $w_{a,c,e}^{l,n,d}$ shows the weight at position a, c, e related to n th feature map in the l th layer. The row, column, and depth sizes of the kernel are shown as a_l, c_l , and e_l , respectively. The 3D convolution has the computational cost of

$$D_k \times D_k \times D_k \times W_b \times H_b \times l_b \times C_b \times C_a \quad (6)$$

where the kernel has a size of $D_k \times D_k \times D_k$ along each dimension. Then, the input and output have dimensions of $W_b \times H_b \times l_b$ and $W_a \times H_a \times l_a$, respectively, where W and H denote the width and height and l denotes the length. C_b and C_a show the number of channels before and after the convolution operation.

In addition, the output of a 2D CNN can be written as below:

$$O_{x,y}^{l,n} = f \left(\sum_d \sum_{a=0}^{a_l-1} \sum_{c=0}^{c_l-1} w_{a,c}^{l,n,d} * O_{x+a,y+b}^{l-1,d} + b^{l,n} \right) \quad (7)$$

where n is the number of feature maps in layer l . $O_{x,y}^{l,n}$ is the output of 2D convolution at position x, y , and channel n which is computed by adding the network bias $b^{l,n}$ to the result of applying the activation function f . The set of output feature maps of layer $l-1$ is considered as d which is the input of layer l . $w_{a,c}^{l,n,d}$ shows the weight at position a, c related to n th feature map in the l th layer. The row and column sizes

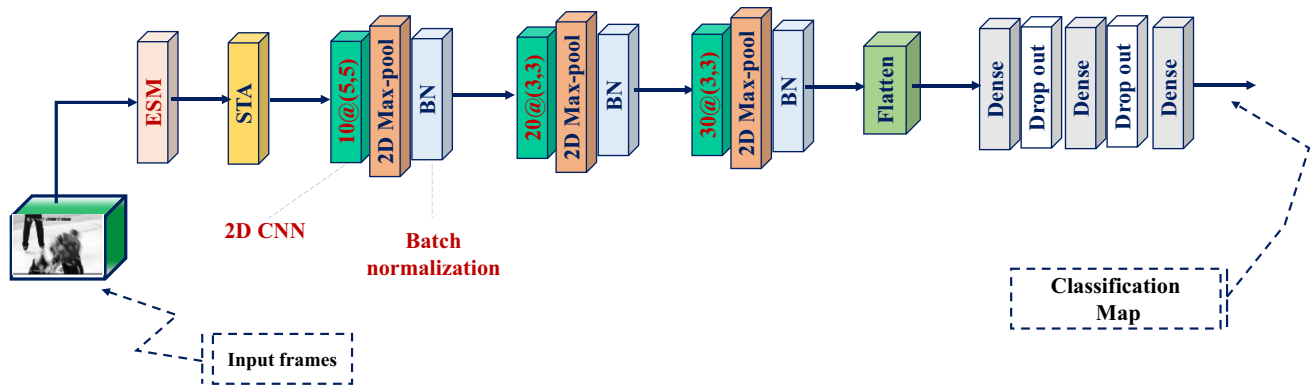


Fig. 9 The proposed framework without feature fusion

of the kernel are shown as a_l and c_l , respectively. The computational cost of the 2D CNN can be formulated as below:

$$D_k \times D_k \times W_b \times H_b \times C_b \times C_a \quad (8)$$

where $D_k \times D_k$ shows the kernel size. $W_b \times H_b$ denotes the input size. C_b and C_a are the number of channels before and after applying the 2D convolution.

From formulas (5)–(8), it is clear that 3D CNNs have an extra dimension for the kernel, the input, and the output, which means they have more parameters and higher complexity than 2D CNNs.

5.5 Ablation studies

In this section, we conducted ablation studies on the Violent Flows and Surveillance Fight datasets. The first ablation study was conducted on feature fusion within the CNN architecture by removing it from the architecture. As a result, the architecture of the 2D CNN was modified according to Fig. 9. The results in Table 5 demonstrate that performance improves by using feature fusion in a 2D CNN architecture. Specifically, on the Violent Flows dataset, the accuracy rate increased from 96.63 to 98.53%. On the Surveillance Fight dataset, the accuracy rate increased from 97.7 to 98.44%.

Another ablation study was performed on the STA block by replacing global average pooling with global max pooling. Table 6 shows that the accuracy of the proposed model is not significantly affected by the global pooling type used in the STA block. Finally, we selected global average pooling because the accuracy rates of the two datasets are rather higher than in the case of global max pooling.

Another ablation study was conducted to evaluate the radius size of the structuring element used in the entropy filter. The results in Table 7 indicate that using a disk with a radius of 9 pixels yields better performance on two datasets. Table 7 shows that the radius size of the structuring element has no significant impact on the accuracy of the

Table 5 Ablation studies for feature fusion in the architecture of the 2D CNN

Module	Violent Flows		Surveillance Fight	
	Accuracy	SD	Accuracy	SD
The proposed framework (Fig. 1)	98.53	6.83	98.44	6.9
The proposed framework without Fusion (Fig. 9)	96.63	8.88	97.7	6.49

Table 6 Ablation studies on the STA block

Module	Violent Flows		Surveillance Fight	
	Accuracy	SD	Accuracy	SD
The STA block with global average-pooling	98.53	6.83	98.44	6.9
The STA block with global max-pooling	98.19	7.92	97.98	7.97

Table 7 Ablation studies on the radius of the structuring element

Radius of disk	Violent Flows		Surveillance Fight	
	Accuracy	SD	Accuracy	SD
5	97.65	8.55	98.22	8.96
7	98.25	7.44	98.16	7.56
9	98.53	6.83	98.44	6.9

Surveillance Fight dataset. However, the radius of the structuring element has a greater impact on the crowded scenes than the uncrowded scenes.

We performed an additional ablation study by removing the ESM from our framework, and we reported the resulting accuracy. The architecture without the ESM is illustrated in

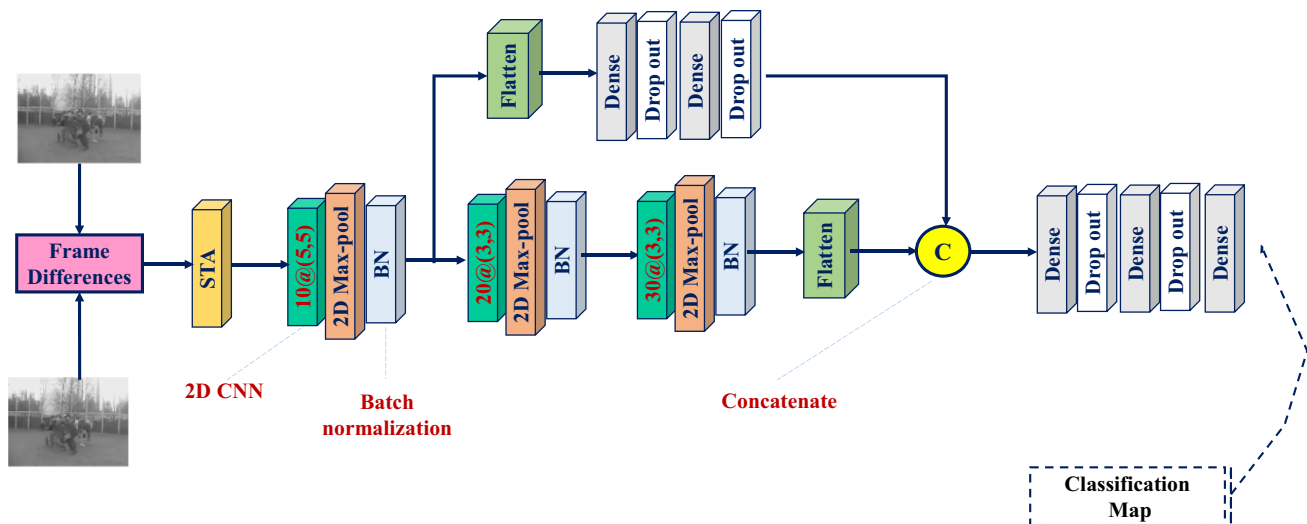


Fig. 10 The proposed architecture without the ESM

Fig. 10. To further evaluate the effectiveness of the ESM, we applied frame differences to the proposed architecture and reported the results in Table 8. The results demonstrate that the ESM effectively increases the accuracy rate from 97.25 to 98.53% on the Violent Flows dataset and from 97.1 to 98.44% on the Surveillance Fight.

6 Conclusion

We proposed spatial and temporal attention as well as a squeeze module to enable a 2D CNN to encode spatiotemporal characteristics. For spatial attention, the ESM was introduced to highlight regions with the potential for violent behaviors. In particular, the ESM used the entropy filter to highlight regions with more movement. For temporal attention, we introduced the STA block and a squeeze part that could summarize a video stream into three channels. Through several experiments, we proved the effectiveness of our suggested modules with an effective 2D CNN, the ESM, and the STA block. We evaluated our proposed method on four benchmark datasets and compared it with several state-of-the-art methods. The experimental results showed that our method achieved superior performance in

terms of accuracy. Our approach achieved high performance on the Surveillance Fight, Action Movies, and Hockey Fight datasets, with an accuracy rate of 98.44%, 100%, and 99.7%, respectively. Similarly, on the Violent Flows dataset, we achieved an accuracy of 98.53%. We also conducted ablation studies to demonstrate the effectiveness and necessity of each component of our method. In the future, we will continue to propose novel architectures to train a more reliable model with fewer parameters compared to 3D CNNs. We will also broaden our research to include a variety of action recognition tasks for a wide range of purposes.

Author contributions Javad Mahmoodi: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing- Original draft preparation, Visualization. Hossein Nezamabadi-pour: Conceptualization, Investigation, Writing-Reviewing and Editing, Supervision, Project Administration.

Data availability The datasets generated during and/or analyzed during the current study are available in the following repositories: **Hockey Fight dataset:** <http://visilab.etsii.uclm.es/personas/oscar/FightDetection/HockeyFights.zip>. **Action Movies dataset:** <http://visilab.etsii.uclm.es/personas/oscar/FightDetection/Películas.rar>. **Violent Flows dataset:** <http://www.openu.ac.il/home/hassner/data/violentflows>. **Surveillance Fight dataset:** <https://github.com/seymenurakti/fight-detection-surv-dataset>.

Code availability Codes have been available at: <https://github.com/javad1359/Violence-Detection>

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 8 Remove the ESM from the proposed architecture

Module	Violent Flows		Surveillance Fight	
	Accuracy	SD	Accuracy	SD
The proposed framework without the ESM (Fig. 10)	97.25	7.72	97.1	8.5
The proposed framework (Fig. 1)	98.53	6.83	98.44	6.9

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